

BUILT — Speed pass, round 2 (v0.57.0)

Four more internal optimizations, each MEASURED. Fewer bytes, less work per request, faster type-ahead, better concurrency during OCR.

☐ Compact JSON

- `json.dumps` now uses `separators=(',', ':')` — no spaces between keys/values.
- Every API response is smaller on the wire (and the smaller text gzips a touch better too).
- Zero behaviour change; purely fewer bytes.

MEASURED: ~8% smaller JSON payloads (then gzipped).

☐ Pre-render page ETag

- The `/page` route computes a cheap param-based ETag and checks `If-None-Match` BEFORE rendering.
- A repeat view returns 304 without opening the PDF, rendering, or hashing the image at all.
- `_send` also stops md5-hashing big PNGs (it trusts the supplied ETag).

MEASURED: 304 on repeat view -> renderer NOT touched (X-Rendered: no).

⚡ Precomputed suggest table

- `optimize_index.py` builds `suggest_terms(term PRIMARY KEY, freq)` from the FTS vocab once.
- Type-ahead becomes a prefix lookup on a WITHOUT-ROWID b-tree instead of a GROUP BY over the whole FTS vocab every keystroke; a small LRU caches recent prefixes.
- Falls back to the vocab GROUP BY until the table is built.

MEASURED: 46x faster per keystroke (0.18ms -> ~0.00ms; bigger on the real corpus).

☐ WAL journal mode

- `optimize_index.py` switches the DB to WAL (Write-Ahead Logging).
- Server reads no longer block on the OCR writer — readers and the single writer run concurrently.
- Reversible (`PRAGMA journal_mode=DELETE`); local-disk only (as designed).

MEASURED: Concurrent reads during writes (verified: `journal_mode=wal`).

EVERY CHANGE MEASURED — AND STILL GREEN

Per your rule, each optimization was verified to actually reduce work or bytes: smaller JSON, a 304 that skips rendering, a 46x type-ahead query, WAL concurrency. All server-internal, RPS-safe, additive. 23 pillars + 21 feature tests still pass (the full module imports cleanly with every edit). `suggest_terms` + WAL are built by `optimize_index.py` — run it once when OCR is paused.

BUILT diagram. Dark (R3). v0.57.0 · 2026-06-02 · `viewer_app.py (_send · /page · suggest)` + `optimize_index.py (suggest_terms · WAL)`. Additive (R1/R6).